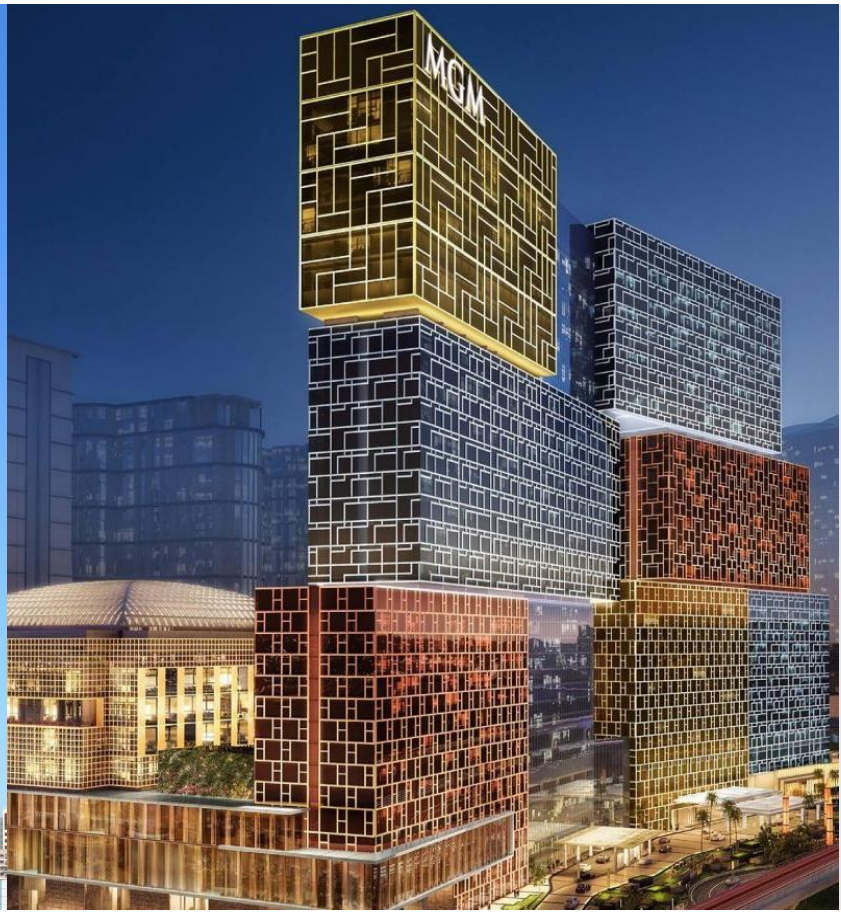
MGM MACAU & COTAI
RESOURCE EFFICIENCY MANAGEMENT REPORT
JUNE 2025

Prepared For:

MGM GRAND PARADISE LIMITED
MGM MACAU, Avenida Dr. Sun Yat Sen Nape, Macau

MGM COTAI, Avenida da Nave Desportiva, Cotai, Macau





NV5 NET ZERO

Unit 5B, 5/F, 133 Wai Yip Street,
Kwun Tong, Hong Kong
Phone: +852 2161 1360

860723-0004415.00
860723-0004367.00

EXECUTIVE SUMMARY

NV5 has been working closely with MGM to implement the EMOs as well as identify additional opportunities to achieve ISO certification energy targets. NV5 appreciates the great support and trust from MGM team over the years, in particular the Facilities Management and the Sustainability Teams.

Key Performance Indicators

NV5 reports both normalized and actual savings on a monthly basis for both properties. The normalized savings are determined based on a 2019 baseline, with aligned annual savings targets to meet both ISO50001 as well as NV5's retainer contract requirements. Please note that the reported normalized savings achieved are to-date and cumulative for each month of the respective year.

Table 1. ISO and Retainer Contract Normalized Energy Performance for 2025

MGM Properties	Baseline	2025 Normalized Energy Target	2025 Normalized Energy Savings
MACAU	2019	-13%	-24.1%
COTAI			-10.8%
Overall			**-16.6%**

The actual monthly savings are determined against the 2019 baseline consumption and cumulative for each month of 2025.

Table 2. Internal Actual Energy Performance for 2025

MGM Properties	Baseline	2025 Internal Energy Target	2025 Actual Energy Savings
MACAU	2019	-13%	+6.6%
COTAI			-1.8%
Overall			**+1.4%**

Highlighted Changes in Operation

MACAU

1. MGM MACAU had turned ON the Main Boiler and two Small Boilers on 10 – 17 Jan 2025, 26 -30 Jan 2025 and 07 – 10 Feb 2025 due to low outdoor temperature.
2. SCHWP-1 & 3 retrofit works were completed. SCHWP-6 retrofit works completed by Q1 2025.
3. MGM MACAU conducted Main Boiler annual maintenance on 15 – 20 May 2025.
4. L35 Villa re-opened on early May 2025.
5. Hotel Lobby had performed LED replacement for 24 pc track mounted spotlight

COTAI

1. All 13 units of PICCV installation works were completed on 23 January 2025.
2. NG Boilers were operated on 20 March 2025 from 10:00 to 14:00 due to quarterly maintenance of Heat Pump CH-4.
3. 3F Mansion Kitchen KEF steps speed control was completed on 15 Apr 2025.

Consumption Intensity

The 2025 YTD electricity and water use intensities are tracked on a monthly basis for both properties:

Table 3. Year to Date Actual Electricity Use Intensity (EUI)

MGM Properties	YTD EUI (kWh/m²)	% Change Compared to 2024
MACAU	**28.3**	**+5.0%**
COTAI	**26.4**	**-2.7%**

Table 4. Year to Date Actual Water Use Intensity (WUI)

MGM Properties	YTD WUI (m³/m²)	% Change Compared to 2024
MACAU	**0.30**	**+8.4%**
COTAI	**0.20**	**+7.4%**

ISO 50001 Certification Audit

MACAU

The HKQAA 2025 surveillance audit was completed on 20 February 2025. There is no nonconformity (NC) and areas for improvement (AFI) from auditors.

COTAI

The HKQAA 2024 surveillance audit was completed on 8 December 2024. There is no nonconformity (NC) and areas for improvement (AFI) from auditors.

Milestones in Year 2025

The following activities will be completed in the year 2025:

Table 5. MGM MACAU Year 2025 Projects

Item	Initiatives	Status
Electricity		
1	Chiller Replacement (Phase 2 : 3 units)	On-going
2	Cooling Tower Fan Retrofit (Phase 3 : 3 units)	On-going
3	AHU/PAU EC Fan Replacement (Phase 2 : 15 units)	On-going
4	Secondary Chilled Water Pump Retrofit (Phase 2 : 1 unit)	On-going

Table 6. MGM COTAI Year 2025 Projects

Item	Initiatives	Status
Electricity		
1	AHU/PAU EC Fan Replacement (Phase 2 : 9 units, Additional : 3 units)	On-going
2	Energy Valve Retrofit for AHU/PAU (Phase 4 : 10 units)	On-going
3	Additional VSD for Exhaust Air Fan (Phase 3 : 13 units)	On-going
4	Axial Fan Retrofit	On-going
5	Solar Thermal Panel (9 units, 18.3 kW System)	On-going

TABLE OF CONTENTS

EXECUTIVE SUMMARY 1

ISO 50001 Certification Audit 2
Milestones in Year 2025 3

1.0 PORTFOLIO PERFORMANCE SUMMARY 6

1.1 Electricity Use Intensity (EUI) 6
1.2 Water Use Intensity (WUI) 7

2.0 MGM MACAU PERFORMANCE 8

2.1 Electricity consumption 8
2.2 Gas Consumption 10
2.3 ISO50001 Energy Consumption Record 12
2.4 Water Consumption 13
2.5 MACAU Year 2025 Projects 15

3.0 MGM COTAI PERFORMANCE 16

3.1 Electricity Consumption 16
3.2 Natural Gas Consumption 19
3.3 ISO50001 Energy Consumption Record 21
3.4 Water Consumption 22
3.5 COTAI Year 2025 Projects 24

APPENDIX A 25

Regression Analysis 25
MGM MACAU 25
MGM COTAI 25

APPENDIX B 26

MACAU Gas Consumption Raw Data 26

ACKNOWLEDGMENT

NV5 has been working with MGM Team as the energy and sustainability manager to achieve energy savings in MGM properties and feed operational intelligence into the design.

The major goal of this project is not only to realize operation energy savings but also to extend the life of equipment, hence reducing CAPEX. MGM has demonstrated continuous commitment to be an environmentally and socially responsible business in the past years. In addition to that NV5 has assisted MGM MACAU and COTAI to achieve several other energy management and sustainability awards and certifications, such as ISO 50001, CEM Energy Savings Award, Macao Green Hotel Award and China 3 Star Green Building Operation Label.

NV5 appreciates the great support and trust from MGM Team over the years, especially the Facilities Management and Sustainability Teams.

1.0 PORTFOLIO PERFORMANCE SUMMARY

1.1 ELECTRICITY USE INTENSITY (EUI)

MACAU's electricity use intensity (EUI) in June 2025 increased by 18.4% compared to the same month in 2024 and increased by 5.0% in 2025 compared to 2024. Meanwhile, COTAI's EUI in June 2025 decreased by 1.7% compared to the same month in 2024 and decreased by 2.7% in 2025 compared to 2024.

Table 7. Actual Electricity Consumption Intensity (EUI) Comparison

Monthly Period	MACAU			COTAI		
	Year 2024 Actual EUI (kWh/m²)	Year 2025 Actual EUI (kWh/m²)	% Change	Year 2024 Actual EUI (kWh/m²)	Year 2025 Actual EUI (kWh/m²)	% Change
January	23.7	24.2	+2.1%	24.2	23.8	-1.5%
February	23.0	22.3	-3.1%	23.0	21.7	-5.6%
March	25.9	25.7	-0.9%	25.9	25.1	-3.0%
April	29.4	27.8	-5.4%	28.9	26.9	-6.9%
May	29.3	33.9	+15.8%	29.5	30.1	+2.1%
June	30.4	36.0	+18.4%	31.5	31.0	-1.7%
July						
August						
September						
October						
November						
December						
Average	**26.9**	**28.3**	**+5.0%**	**27.2**	**26.4**	**-2.7%**

1.2 WATER USE INTENSITY (WUI)

MACAU's water use intensity (WUI) in June 2025 increased by **9.6%** compared to the same month in 2024 and increased by **8.4%** in 2025 compared to 2024. Meanwhile, COTAI's WUI in June 2025 increased by **6.7%** compared to the same month in 2024 and increased by **7.4%** in 2025 compared to 2024.

Table 8. Actual Water Consumption Intensity (WUI) Comparison

Monthly Period	MACAU			COTAI		
	Year 2024 Actual WUI (m^3/m^2)	Year 2025 Actual WUI (m^3/m^2)	% Change	Year 2024 Actual WUI (m^3/m^2)	Year 2025 Actual WUI (m^3/m^2)	% Change
January	0.239	0.250	+4.7%	0.172	0.173	+0.8%
February	0.257	0.293	+14.1%	0.179	0.197	+10.3%
March	0.270	0.292	+7.9%	0.175	0.198	+13.0%
April	0.316	0.310	-2.1%	0.212	0.216	+1.9%
May	0.307	0.359	+16.8%	0.204	0.228	+11.9%
June	0.297	0.325	+9.6%	0.200	0.213	+6.7%
July						
August						
September						
October						
November						
December						
Average	**0.281**	**0.305**	**+8.4%**	**0.190**	**0.204**	**+7.4%**

2.0 MGM MACAU PERFORMANCE

2.1 ELECTRICITY CONSUMPTION

The actual electricity consumption and cost in June 2025 increased by **18.4%** and **13.9%**, respectively, compared to the same month in 2024. To date, MGM MACAU has increased the actual electricity consumption by **5.0%** and associated costs increased by **1.6%.**

Table 9. MACAU Summary of Actual Monthly Electricity Consumption and Cost

Monthly Period	Actual Electricity Consumption (kWh)			Actual Electricity Cost (MOP)			% Change – CDD
	2024	2025	% Change Actual	2024	2025	% Change Actual	
January	5,003,520	5,110,080	+2.1%	5,973,270	5,893,310	-1.3%	-61.2%
February	4,860,480	4,708,680	-3.1%	5,812,030	5,448,990	-6.2%	-57.4%
March	5,485,440	5,434,200	-0.9%	6,530,510	6,282,750	-3.8%	-21.8%
April	6,220,200	5,881,200	-5.4%	7,458,080	6,787,570	-9.0%	-31.0%
May	6,200,040	7,176,840	+15.8%	7,425,270	8,264,780	+11.3%	+11.7%
June	6,425,760	7,607,760	+18.4%	8,437,580	9,613,640	+13.9%	+2.8%
July							
August							
September							
October							
November							
December							
Total	**34,195,440**	**35,918,760**	**+5.0%**	**41,636,740**	**42,291,040**	**+1.6%**	**-12.7%**

***Important note**: Cooling Degree Days (CDDs) and food cover are the main variables taken into consideration for electricity consumption normalization.*

NV5 summarized the key items that could result in an electricity consumption increase below:

- Chiller Plant electricity consumption increased by 17.1% in June 2025 compared with the same month in 2024, as CDD increase 2.8%.
- Air Side Equipment electricity consumption increased by 12.0% in June 2025 compared with the same month in 2024.
- Art Museum and Fantasy Box consumed 79,746 kWh and 172,229 kWh electricity respectively in June 2025.

Table 10 presents a comparison of electricity rates in the years 2024 and 2025. The electricity rate in June 2025 decreased by 3.8% compared to the same month in 2024.

Table 10. MACAU - Summary of Actual Monthly Electricity Rate

Monthly Period	Year 2024 – Electricity Rate (MOP/kWh)	Year 2025 – Electricity Rate (MOP/kWh)	% Change – Electricity Rate
January	1.194	1.153	-3.4%
February	1.196	1.157	-3.2%
March	1.191	1.156	-2.9%
April	1.199	1.154	-3.7%
May	1.198	1.152	-3.8%
June	1.313	1.264	-3.8%
July			
August			
September			
October			
November			
December			
Average	**1.215**	**1.173**	**-3.5%**

Table 11 summarizes the monthly normalized electricity consumption. The actual electricity consumption in June 2025 decreased by 17.4% compared to 2019 normalized consumption in the same month.

Table 11. MACAU - Summary of Normalized Monthly Electricity Consumption

Monthly Period	2019 Normalized Electricity Consumption (kWh)	2025 Actual Electricity Consumption (kWh)	% Change – Normalized
January	7,507,180	5,110,080	-31.9%
February	7,133,797	4,708,680	-34.0%
March	7,590,986	5,434,200	-28.4%
April	7,955,390	5,881,200	-26.1%
May	9,380,208	7,176,840	-23.5%
June	9,205,733	7,607,760	-17.4%
July			
August			
September			
October			
November			
December			
Total	**48,773,295**	**35,918,760**	**-26.4%**

2.2 GAS CONSUMPTION

Gas consumption can be broken down into three (3) categories: boiler, hotel upper zone DHW, and kitchens/restaurants. Since the boiler and restaurants Gas utilization covers over 95% of the overall site Gas consumption, NV5 focuses the analysis on those two areas.

As of 16 January 2024, MGM Macau has transitioned from LPG to Natural Gas for all gas-operated equipment.

According to Table 12, the actual Gas consumption in June 2025 increased by **15.3%** while the cost increased by **11.5%** respectively compared to the same month in 2024. To date, MGM MACAU has decreased the actual Gas consumption by **4.3%** and associated costs decreased by **6.7%**.

Table 12. MACAU – Summary of Actual Monthly Gas Consumption and Cost

Monthly Period	Actual Gas Consumption (m^3)			Actual Gas Cost (MOP)			% Change – Min OAT	% Change – Food Covers
	2024 (LPG + NG)	2025 (NG)	% Change Actual	2024 (LPG + NG)	2025 (NG)	% Change Actual		
January	165,990*	144,889	-12.7%	1,211,135	964,330	-20.4%	+101.6%	+34.2%
February	158,122	127,213	-19.5%	1,052,400	846,670	-19.5%	+34.7%	+23.7%
March	133,119	125,338	-5.8%	885,980	834,200	-5.8%	+13.4%	+19.9%
April	117,800	116,979	-0.7%	784,030	778,570	-0.7%	-32.6%	+21.6%
May	108,947	121,147	+11.2%	725,110	806,310	+11.2%	+4.1%	+25.8%
June	97,924	112,902	+15.3%	651,740	726,570	+11.5%	+5.5%	+22.1%
July								
August								
September								
October								
November								
December								
Total	**781,902**	**748,468**	**-4.3%**	**5,310,395**	**4,956,649**	**-6.7%**	**+6.7%**	**+24.5%**

Important note: *Minimum Outdoor Air Temperature (OAT) is the main variable taken into consideration for Gas consumption normalization.*
**Jan 2024 gas consumption was combined with LPG & NG, it was converted as equivalent Gas (m^3) for comparison.*

NV5 summarized the key items which could result in Gas consumption increase below:

- Kitchen Gas consumption increased by 53.6% in June 2025 as compared with the same month in 2024.

Table 13 presents the comparison of Gas rates in the years 2024 and 2025. The gas rate in June 2025 decreased by 3.3% compared to the same month in 2024.

Table 13. MACAU - Summary of Actual Monthly Gas Rate

Monthly Period	Year 2024 – Gas Rate (MOP/m³)	Year 2025 – Gas Rate (MOP/m³)	% Change – Gas Rate
January	7.30	6.66	-8.8%
February	6.66	6.66	-
March	6.66	6.66	-
April	6.66	6.66	-
May	6.66	6.66	-
June	6.66	6.44	-3.3%
July			
August			
September			
October			
November			
December			
Average	**6.76**	**6.62**	**-2.1%**

*Jan 2024 gas rate was combined with LPG & NG, it was converted as equivalent Gas (m³) for comparison

Table 14 summarizes the monthly normalized Gas consumption. The actual Gas consumption in June 2025 has decreased by 11.6% compared to 2019 normalized consumption in the same month.

Table 14. MACAU - Summary of Normalized Monthly Gas Consumption

Monthly Period	2019 Normalized Gas Consumption (m³)	2025 Actual Gas Consumption (m³)	% Change – Normalized
January	145,796	144,889	-0.6%
February	142,968	127,213	-11.0%
March	160,521	125,338	-21.9%
April	134,756	116,979	-13.2%
May	126,081	121,147	-3.9%
June	127,765	112,902	-11.6%
July			
August			
September			
October			
November			
December			
Total	**837,887**	**748,468**	**-10.7%**

2.3 ISO50001 ENERGY CONSUMPTION RECORD

The MACAU ISO50001 energy performance target is to achieve 13% normalized energy savings by 2025 compared to the 2019 baseline. To date, normalized energy saving achieved up to June 2025 is **24.1%** as shown in Table 15. Table 16 below summarizes the site monthly energy consumption (Electricity and NG combined) compared to the 2019 baseline.

Table 15. MACAU ISO 50001 Normalized Energy Performance Targets for 2025

MGM Properties	Baseline	Normalized Savings	
MACAU	2019	2025 Target	**-13%**
		Savings as of June 2025	**-24.1%**

Table 16. MACAU ISO 50001 Normalized Monthly Energy Consumption

Monthly Period	2019 Normalized Energy Consumption (GJ)	2025 Actual Energy Consumption (GJ)	% Change – Normalized
January	32,085	23,424	-27.0%
February	30,643	21,366	-30.3%
March	32,898	23,912	-27.3%
April	33,315	25,231	-24.3%
May	38,144	30,040	-21.2%
June	37,574	31,306	-16.7%
July			
August			
September			
October			
November			
December			
Total	**204,659**	**155,279**	**-24.1%**

Important note: *Factors used for unit conversion: Electricity (0.0036 GJ/kWh), NG (0.0347 GJ/m³)*

2.4 WATER CONSUMPTION

Based on the existing water metering system, the overall site water consumption can be broken down into five (5) categories: Kitchen and Restaurants; Hotel Tower; Fire Services; Chiller Plant & Miscellaneous.

According to Table 17, the actual water consumption and cost in June 2025 increased by 9.6% and 9.6% respectively compared to the same month in 2024 respectively. To date, MGM MACAU has increased both the actual water consumption and associated cost by 8.4% and 8.4% respectively.

Table 17. MACAU - Summary of Actual Monthly Water Consumption and Cost

Monthly Period	Actual Water Consumption (m^3)			Actual Water Cost (MOP)			Normalization Factors	
	2024	2025	% Change Actual	2024	2025	% Change Actual Cost	% Change – CDD	% Change – Food Covers
January	50,588	52,988	+4.7%	392,330	410,980	+4.8%	-61.2%	+34.2%
February	54,323	61,961	+14.1%	421,370	480,670	+14.1%	-57.4%	+23.7%
March	57,148	61,675	+7.9%	443,270	478,430	+7.9%	-21.8%	+19.9%
April	66,952	65,531	-2.1%	519,260	508,320	-2.1%	-31.0%	+21.6%
May	65,014	75,919	+16.8%	504,150	588,810	+16.8%	+11.7%	+25.8%
June	62,753	68,780	+9.6%	486,660	533,440	+9.6%	+2.8%	+22.1%
July								
August								
September								
October								
November								
December								
Total	**356,778**	**386,854**	**+8.4%**	**2,767,040**	**3,000,650**	**+8.4%**	**-12.7%**	**+24.5%**

Important note: *Cooling Degree Days (CDDs) and Food Covers are the main variables taken into consideration for water consumption normalization.*

NV5 summarized the key items that could result in a water consumption increase below:

- Kitchen water consumption has increased by 17.0% compared with the same month in 2024, as Food Cover increased 22.1%.
- CT Make Up water consumption has increased by 42.8% compared with the same month in 2024, as CDD increased by 2.8%.
- Hotel water consumption has increased by 10.4% compared with the same month in 2024.

Table 18 presents the comparison of water rates in the years 2024 and 2025. The water rate in June 2025 increased by 0.01% compared to the same month in 2024. The average water rate in 2025 increased by 0.01% compared to 2024.

Table 18. MACAU - Summary of Actual Monthly Water Rates

Monthly Period	Year 2024 – Water Rate (MOP/m³)	Year 2025 – Water Rate (MOP/m³)	% Change – Water Rate
January	7.7554	7.7561	+0.01%
February	7.7568	7.7576	+0.01%
March	7.7565	7.7573	+0.01%
April	7.7557	7.7569	+0.02%
May	7.7545	7.7558	+0.02%
June	7.7552	7.7557	+0.01%
July			
August			
September			
October			
November			
December			
Average	**7.7557**	**7.7566**	**+0.01%**

Table 19 summarized the monthly normalized water consumption. The actual water consumption in June 2025 decreased by 13.9% compared to 2019 normalized consumption in the same month.

Table 19. MACAU - Summary of Normalized Monthly Water Consumption

Monthly Period	2019 Normalized Water Consumption (m³)	2025 Actual Water Consumption (m³)	% Change – Normalized
January	68,054	52,988	-22.1%
February	64,760	61,961	-4.3%
March	68,070	61,675	-9.4%
April	70,467	65,531	-7.0%
May	81,837	75,919	-7.2%
June	79,925	68,780	-13.9%
July			
August			
September			
October			
November			
December			
Total	**433,114**	**386,854**	**-10.7%**

2.5 MACAU YEAR 2025 PROJECTS

NV5 continues to perform analysis and coordinate with MGM to implement the following projects in 2025 and reduce the site energy consumption which supports MGM energy savings target.

Table 20. MACAU Year 2025 Projects

Item	Initiatives	Status
Electricity		
1	Chiller Replacement (Phase 2 : 3 units)	On-going
2	Cooling Tower Fan Retrofit (Phase 3 : 3 units)	On-going
3	AHU/PAU EC Fan Replacement (Phase 2 : 15 units)	On-going
4	Secondary Chilled Water Pump Retrofit (Phase 2 : 1 unit)	On-going

3.0 MGM COTAI PERFORMANCE

3.1 ELECTRICITY CONSUMPTION

The actual electricity consumption in June 2025 decreased by 1.7% and actual electricity cost decreased by 5.6% compared to the same month in 2024. To date, MGM COTAI has decreased the actual electricity consumption by 2.7% and associated costs decreased by 6.1%.

Table 21. COTAI Summary of Actual Monthly Electricity Consumption and Cost

Monthly Period	Actual Electricity Consumption (kWh)			Actual Electricity Cost (MOP)			% Change – CDD
	2024	2025	% Change Actual	2024	2025	% Change Actual	
January	9,054,720	8,917,594	-1.5%	11,239,000	10,716,410	-4.6%	-61.2%
February	8,597,400	8,119,200	-5.6%	10,692,600	9,789,270	-8.4%	-57.4%
March	9,700,320	9,408,030	-3.0%	12,045,940	11,322,600	-6.0%	-21.8%
April	10,820,760	10,071,744	-6.9%	13,530,960	12,120,910	-10.4%	-31.0%
May	11,032,920	11,268,384	+2.1%	13,782,190	13,524,230	-1.9%	+11.7%
June	11,777,640	11,579,720	-1.7%	14,717,760	13,891,430	-5.6%	+2.8%
July							
August							
September							
October							
November							
December							
Total	**60,983,760**	**59,364,672**	**-2.7%**	**76,008,450**	**71,364,850**	**-6.1%**	**-12.7%**

***Important note**: Cooling Degree Days (CDDs), Food Cover, Room Night and E-Bus Mileage are the main variables taken into consideration on electricity consumption normalization. Starting from Jan 2025, LRT Interface electricity consumption is included.*

NV5 summarized the key items which could result in an electricity consumption decrease below:

- Hotel Tower electricity consumption decreased by 13.5% in June 2025 compared with the same month in 2024, North Tower Tier 3 is under major renovation.
- Heat Pump Plant electricity consumption decreased by 6.2% in June 2025 compared with the same month in 2024.

Table 22 presents the monthly e-bus electricity consumption in MGM COTAI. E-bus electricity consumption in June 2025 increased by 2.6% compared to the same month in 2024.

Table 22. COTAI - Summary of Monthly e-bus Electricity Consumption

Monthly Period	E-bus Electricity Consumption (kWh)		
	2024	2025	% Change – Actual
January	137,030	145,776	+6.4%
February	142,468	140,426	-1.4%
March	151,047	160,250	+6.1%
April	162,250	153,506	-5.4%
May	163,600	175,131	+7.0%
June	170,502	174,993	+2.6%
July			
August			
September			
October			
November			
December			
Total	**926,897**	**950,082**	**+2.5%**

Table 23 presents the comparison of electricity rates in the years 2024 and 2025. The electricity rate in June 2025 decreased by 4.0% compared to the same month in 2024.

Table 23. COTAI - Summary of Actual Monthly Electricity Rate

Monthly Period	Year 2024 – Electricity Rate (MOP/kWh)	Year 2025 – Electricity Rate (MOP/kWh)	% Change – Electricity Rate
January	1.241	1.202	-3.2%
February	1.244	1.206	-3.1%
March	1.242	1.203	-3.1%
April	1.250	1.203	-3.8%
May	1.249	1.200	-3.9%
June	1.250	1.199	-4.0%
July			
August			
September			
October			
November			
December			
Average	**1.245**	**1.203**	**-3.4%**

Table 24 summarizes the monthly normalized electricity consumption. The actual electricity consumption in June 2025 decreased by 9.2% compared to 2019 normalized consumption in the same month.

Table 24. COTAI - Summary of Normalized Monthly Electricity Consumption

Monthly Period	2019 Normalized Electricity Consumption (kWh)	2025 Actual Electricity Consumption (kWh)	% Change – Normalized
January	9,460,497	8,917,594	-5.7%
February	9,413,041	8,119,200	-13.7%
March	10,261,601	9,408,030	-8.3%
April	11,077,435	10,071,744	-9.1%
May	12,394,709	11,268,384	-9.1%
June	12,747,080	11,579,720	-9.2%
July			
August			
September			
October			
November			
December			
Total	**65,353,383**	**59,364,672**	**-9.2%**

3.2 NATURAL GAS CONSUMPTION

The actual natural gas consumption in June 2025 increased by **5.9%** while the cost increased by **2.4%** compared to the same month in 2024, respectively. To date, MGM COTAI has increased actual natural gas consumption by **8.5%** and associated costs increased by **7.2%.**

Table 25. COTAI - Summary of Actual Monthly Natural Gas Consumption and Cost

Monthly Period	Actual NG Consumption (m^3)			Actual NG Cost (MOP)			% Change – Min OAT	% Change – Food Covers
	2024	2025	% Change Actual	2024	2025	% Change Actual		
January	88,198	89,215	+1.2%	607,700	593,780	-2.3%	+101.6%	+65.0%
February	74,893	83,343	+11.3%	498,460	554,700	+11.3%	+34.7%	+91.7%
March	76,680	91,400	+19.2%	510,350	608,320	+19.2%	+13.4%	+82.5%
April	77,305	85,334	+10.4%	514,510	567,950	+10.4%	-32.6%	+79.3%
May	84,651	88,780	+4.9%	563,400	590,890	+4.9%	+4.1%	+78.7%
June	82,348	87,195	+5.9%	548,080	561,140	+2.4%	+5.5%	+75.9%
July								
August								
September								
October								
November								
December								
Total	**484,075**	**525,267**	**+8.5%**	**3,242,500**	**3,476,780**	**+7.2%**	**+6.7%**	**+78.6%**

Important note: *Minimum Outdoor Air Temperature is the main variable taken into consideration on natural gas consumption normalization*

NV5 summarized the key items which could result in natural gas consumption increase below:

- Kitchen NG consumption for June 2025 has increased by 15.2% and food cover has increased by 75.9% compared to the same month in 2024.

Table 26 presents the comparison of natural gas rates in the years 2024 and 2025. The Gas rates in June 2025 decreased by 3.3% compared to the same month in 2024.

Table 26. COTAI - Summary of Actual Monthly Natural Gas Rate

Monthly Period	Year 2024 – Natural Gas Rate (MOP/m³)	Year 2025 – Natural Gas Rate (MOP/m³)	% Change – NG Rate
January	6.89	6.66	-3.4%
February	6.66	6.66	-
March	6.66	6.66	-
April	6.66	6.66	-
May	6.66	6.66	-
June	6.66	6.44	-3.3%
July			
August			
September			
October			
November			
December			
Average	**6.69**	**6.62**	**-1.1%**

Table 27 summarizes the monthly normalized natural gas consumption. The actual natural gas consumption in June 2025 decreased by 16.9% compared to 2019 normalized consumption in the same month.

Table 27. COTAI - Summary of Normalized Monthly Natural Gas Consumption

Monthly Period	2019 Normalized Natural Gas Consumption (m³)	2025 Actual Natural Gas Consumption (m³)	% Change – Normalized
January	124,095	89,215	-28.1%
February	123,938	83,343	-32.8%
March	126,762	91,400	-27.9%
April	121,272	85,334	-29.6%
May	108,094	88,780	-17.9%
June	104,957	87,195	-16.9%
July			
August			
September			
October			
November			
December			
Total	**709,118**	**525,267**	**-25.9%**

3.3 ISO50001 ENERGY CONSUMPTION RECORD

COTAI ISO50001 energy target is to achieve 13% normalized energy consumption reduction by 2025 compared to the 2019 baseline. To date, normalized energy saving achieved up to June 2025 is **10.8%** as shown in Table 28. Table 29 below summarizes the site monthly energy consumption (Electricity and Natural Gas combined) compared to the 2019 baseline.

Table 28. COTAI ISO 50001 Normalized Energy Performance Targets for 2025

MGM Properties	Baseline	Normalized Savings	
COTAI	2019	2025 Target	**-13%**
		Savings as of June 2025	**-10.8%**

Table 29. COTAI ISO 50001 Normalized Monthly Energy Consumption

Monthly Period	2019 Normalized Energy Consumption (GJ)	2025 Actual Energy Consumption (GJ)	% Change – Normalized
January	38,364	35,199	-8.2%
February	38,188	32,121	-15.9%
March	41,340	37,040	-10.4%
April	44,087	39,219	-11.0%
May	48,372	43,647	-9.8%
June	49,531	44,713	-9.7%
July			
August			
September			
October			
November			
December			
Total	**259,879**	**231,940**	**-10.8%**

***Important note**: Factors used for unit conversion: Electricity (0.0036 GJ/kWh) and Natural Gas (0.0347 GJ/m³)*

3.4 WATER CONSUMPTION

According to Table 30, the actual water consumption and associated costs in June 2025 have increased by **6.7%** and **6.7%** respectively compared to the same month in 2024 respectively. To date, MGM COTAI has increased both the actual water consumption and associated cost by **7.4%** and **7.4%** respectively.

Table 30. COTAI - Summary of Actual Monthly Water Consumption and Cost

Monthly Period	Actual Water Consumption (m^3)			Actual Water Cost (MOP)			Normalization Factors	
	2024	2025	% Change Actual	2024	2025	% Change Actual Cost	% Change – CDD	% Change – Food Covers
January	64,264	64,806	+0.8%	499,180	503,350	+0.8%	-61.2%	+65.0%
February	66,870	73,743	+10.3%	519,440	572,760	+10.3%	-57.4%	+91.7%
March	65,464	73,985	+13.0%	508,610	574,630	+13.0%	-21.8%	+82.5%
April	79,200	80,700	+1.9%	615,060	626,580	+1.9%	-31.0%	+79.3%
May	76,182	85,229	+11.9%	591,560	661,710	+11.9%	+11.7%	+78.7%
June	74,796	79,832	+6.7%	580,830	619,820	+6.7%	+2.8%	+75.9%
July								
August								
September								
October								
November								
December								
Total	**426,776**	**458,295**	**+7.4%**	**3,314,680**	**3,558,850**	**+7.4%**	**-12.7%**	**+78.6%**

Important note: *Cooling Degree Days (CDDs) and Food Covers are the main variables taken into consideration on Water consumption normalization.*

NV5 summarized the key items which could result in water consumption increase below:

- CT Make Up water consumption for June 2025 increased by 14.0% compared with the same month in 2024.
- Kitchen water consumption for June 2025 increased by 34.7% compared with the same month in 2024, Food Cover increased 75.9%.
- Mansion water consumption for June 2025 increased by 27.6% compared with the same month in 2024.

Table 31 presents the comparison of water rates in the years 2024 and 2025. The water rate in June 2025 has decreased by 0.02% when compared to the same month in 2024. The average water rate decreased by 0.02% when compared to 2024.

Table 31. COTAI - Summary of Actual Monthly Water Rates

Monthly Period	Year 2024 – Water Rate (MOP/m³)	Year 2025 – Water Rate (MOP/m³)	% Change – Water Rate
January	7.768	7.767	-0.01%
February	7.768	7.767	-0.01%
March	7.769	7.767	-0.03%
April	7.766	7.764	-0.02%
May	7.765	7.764	-0.02%
June	7.766	7.764	-0.02%
July			
August			
September			
October			
November			
December			
Average	**7.767**	**7.766**	**-0.02%**

Table 32 summarized the monthly normalized water consumption. The actual water consumption in June 2025 decreased by 33.2% compared to the same month in 2019. To date, the average utility rate of water has decreased by 28.5% compared to 2019.

Table 32. COTAI - Summary of Normalized Monthly Water Consumption

Monthly Period	2019 Normalized Water Consumption (m³)	2025 Actual Water Consumption (m³)	% Change – Normalized
January	95,412	64,806	-32.1%
February	97,606	73,743	-24.4%
March	103,255	73,985	-28.3%
April	106,217	80,700	-24.0%
May	119,072	85,229	-28.4%
June	119,535	79,832	-33.2%
July			
August			
September			
October			
November			
December			
Total	**641,096**	**458,295**	**-28.5%**

3.5 COTAI YEAR 2025 PROJECTS

NV5 continues to perform analysis and coordinate with MGM to implement the following projects in 2025 and reduce the site energy consumption which supports MGM energy savings target.

Table 33. COTAI Year 2025 Projects

Item	Initiatives	Status
Electricity		
1	AHU/PAU EC Fan Replacement (Phase 2 : 9 units, Additional : 3 units)	On-going
2	Energy Valve Retrofit for AHU/PAU (Phase 4 : 10 units)	On-going
3	Additional VSD for Exhaust Air Fan (Phase 3 : 13 units)	On-going
4	Axial Fan Retrofit	On-going
5	Solar Thermal Panel (9 units, 18.3 kW System)	On-going

APPENDIX A

Regression Analysis

The 2019 normalized consumption modelling has remained used for tracking the agreed normalized energy savings target for MGM MACAU and MGM COTAI. The following tables show a summary of the selected independent variables for each utility which are used for the normalized consumption modelling in 2025.

MGM MACAU

Utilities	Variables	
Electricity (kWh)	CDD (Base T=17°C) & Food Cover	
LPG (kg) / Natural Gas (m^3)	Min OAT	
Water (m^3)	CDD (Base T=17°C)	Food Cover

MGM COTAI

Utilities	Variables	
Electricity (kWh)	CDD (Base T=17°C), Food Cover, Room Night, E-Bus Mileage	
Natural Gas (m^3)	Min OAT	
Water (m^3)	CDD (Base T=17°C)	Food Cover

APPENDIX B

MACAU Gas Consumption Raw Data

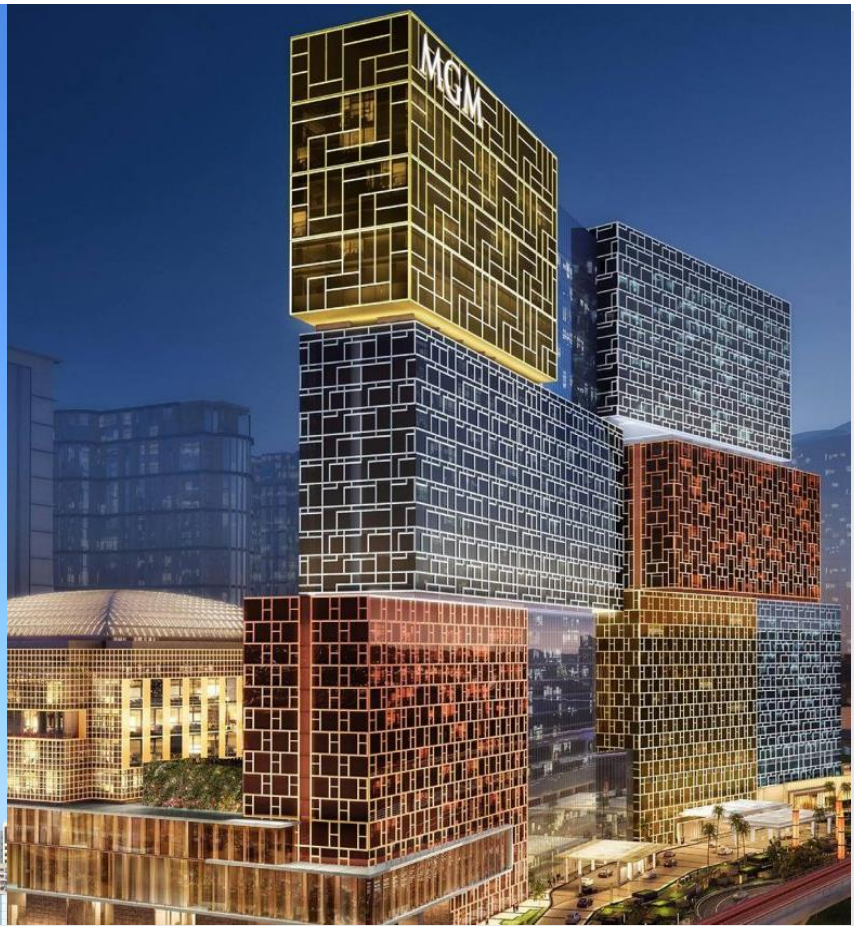
Table 34. MACAU Gas Consumption Raw Data

Monthly Period	2024 LPG Consumption (kg)	2024 NG Consumption (m^3)	2025 NG Consumption (m^3)
January	50,003	100,712	144,889
February	-	158,122	127,213
March	-	133,119	125,338
April	-	117,800	116,979
May	-	108,947	121,147
June	-	97,924	112,902
July	-	99,596	
August	-	101,187	
September	-	101,015	
October	-	107,043	
November	-	108,030	
December	-	119,904	
Total	**50,003**	**1,353,399**	**748,468**

Important note: *Factors used for unit conversion: LPG (0.0453 GJ/kg), NG (0.0347 GJ/m^3)*

MGM MACAU & COTAI
Resource Efficiency Management Report
860723-0004415.00
860723-0004367.00





NV5 NET ZERO

Generating Great Returns. Improving the Environment.